

Deep Space Navigation using X-ray Pulsars

A project report submitted by

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Certificate

This is to certify that Mr. Amoy Ashesh has successfully completed the project titled "*Deep Space Navigation Using X-ray Pulsars*". The project was carried out as part of the internship in the Department of Avionics at the Indian Institute of Space Science and Technology, Thiruvananthapuram.

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Abstract

Humans are driven by a deep desire to explore the unknown. This has led us to uncover vast amounts of information about the universe, from stars and galaxies to black holes. But our thirst for knowledge is far from satisfied. To venture deeper into space and unravel the mysteries of distant planets, we need a new tool: a solar system positioning system (similar to GPS but for space). Ideally, we could even develop a galactic positioning system for missions beyond our solar system. This report explores the development of an autonomous navigation system for spacecraft. This system would reduce or eliminate reliance on communication with Earth for navigation. The first step involves reviewing existing research on how such a system might function.

The report then details how data from NASA's NICER instrument on the ISS was used to create pulse profiles of specific pulsars (spinning neutron stars) that could be used for navigation.

The report describes the software used to process this data, including NASA's HEASoft, shell scripts, and Python. A sensor design is also proposed, along with considerations for its angular resolution and performance.

Finally, the report briefly outlines future work on building a testbed for this navigation system.

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Abbreviations

GPS	Global Positioning System
NASA	National Aeronautics and Space Administration
NICER	Neutron star Interior Composition ExploreR
ISS	International Space Station
HEASoft	High Energy Astrophysics Software
NANOGRAV	North American Nanohertz Observatory for Gravitational Waves
PINT	PINT Is Not TEMPO3
FITS	Flexible Image Transport System
EKF	Extended Kalman Filter
GSO	Geo-Synchronous Orbit
DSN	Deep Space Network
XPNAV	X-ray Pulsar NAVigation
RADAR	RADio Detection And Ranging
INS	Inertial Navigation System
GNSS	Global Navigation Satellite Systems
JPL	Jet Propulsion Laboratory
RXTE	Rossi X-ray Timing Experiment
XMM-Newton	X-ray Multi-Mirror-Newton
MSP	Milli-Second Pulsar
XNAV	X-ray pulsar NAVigation
CCD	Charge-Coupled Device
SDD	Silicon Drift Detector
APS	Active Pixel Sensor
IXO	International X-ray Observatory
SSB	Solar System Barycentre
RA	Right Ascension
DEC	Declination

Chapter 1

Literature Review

X-ray navigation using pulsars, also known as X-ray Pulsar-Based Navigation (XNAV), is an emerging technology that aims to provide autonomous spacecraft navigation by utilizing the periodic X-ray emissions from pulsars. This method offers a promising alternative to traditional Earth-based navigation systems, which are limited by the need for constant communication with ground stations. [1]

1.1 Pulsar Characteristics and X-ray Emissions

Pulsars are highly magnetized, rotating neutron stars that emit beams of electromagnetic radiation. Due to their stable rotation periods, pulsars serve as natural cosmic lighthouses. The X-ray emissions, in particular, provide a reliable signal that can be detected and timed with high precision. According to Becker et al. (2013), X-ray pulsars have distinct advantages over radio pulsars for navigation due to their stronger signals and reduced interference from cosmic noise. [2]

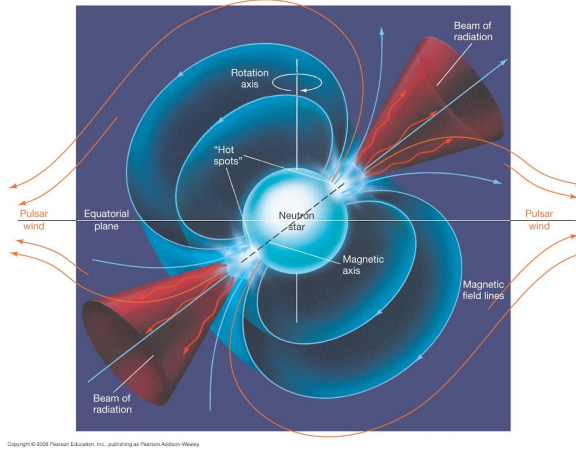


Figure 1.1: The image shows a neutron star emitting beams of X-ray radiation from its magnetic poles, with these beams sweeping across space as the star rotates.

1.2 Theory of navigation using X-ray pulsars

X-ray Pulsar-Based Navigation was initially proposed by Chester and Butman in 1981 [3]. The fundamental principle of X-ray pulsar navigation involves using the concept of time delay. The common approach is to recursively predict the time delay (t_d), which can be calculated from photon Times of Arrival (TOAs). This time delay is directly proportional to the relative distance between the spacecraft and the Solar System Barycenter (SSB), where the reference template is located. This is illustrated in Figure 1.2. The mathematical relation is shown in Equation (1.1).

$$\Delta d = n \cdot \Delta x = ct_d \quad (1.1)$$

This relation only provides the relative distance in one direction. To obtain a complete state measurement, the spacecraft must measure time delays from multiple pulsars in different directions. These measurements can then be input into a Kalman Filter to accurately estimate the spacecraft's position.

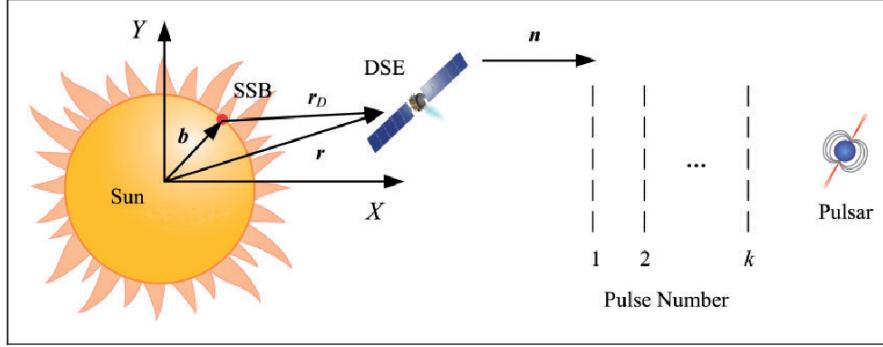


Figure 1.2: A schematic showing the position of the spacecraft, SSB and pulsar direction (Zheng and Wang, 2020)

To sum up the above explanation on navigation structure, this navigation system can be divided into three main steps:[4]

1. Using X-ray detectors to detect pulsar signals.
2. Perform Pulse time delay estimation.
3. Use a Kalman filter to obtain the state estimate.

1.3 NICER/SEXTANT

The Neutron star Interior Composition ExploreR (NICER) is an instrument on the ISS that studies neutron star interiors, providing unprecedented detail on their physics. NICER uses highly precise timing (~ 100 ns resolution) in the soft X-ray range (0.1-12 keV). Its main component, the X-ray Timing Instrument (XTI), employs grazing incidence optics to focus light from neutron stars onto 56 Focal Plane Modules (FPMs). Unlike typical X-ray telescopes, the XTI uses a single mirror assembly for spectral and timing data rather than imaging. Each FPM contains a commercial SDD from Amptek and is grouped in sets of eight into seven Measurement/Power Units (MPUs). Each MPU is independently powered and their data is processed separately. [5]

Chapter 2

General Methodology

2.1 Source Selection for X-ray Pulsar Navigation:

- Determine visibility to the satellite wrt to Earth/celestial body [occulating body]:

$$|\pi - \arccos(n \cdot r_{SC/B})| \leq \arccos \left(\frac{\sqrt{(r_{SC/B})^2 - (R_{SC/B})^2}}{r_{SC/B}} \right) \quad (2.1)$$

- Determine the “good” observing times:

$$\text{Sun angle} > 30^\circ$$

- Query out X-ray sources within 0.5° , 1° , and 6° of the pulsars.
- Determine the score using:

$$\text{Score} = 50(\mathcal{F}_{\text{vis}} + \mathcal{F}_{\text{Sun-source}}) - 20 \cdot \mathcal{N}_{>10} - \frac{\mathcal{N}_{>0.01}}{50} \quad (2.2)$$

2.2 Constraining Detector Specifications

- SNR > 5
- SNR is given by:

$$\begin{aligned}
 S &= \frac{N_{S_{\text{pulsed}}}}{\sigma_{\text{noise}}} \\
 &= \frac{N_{S_{\text{pulsed}}}}{\sqrt{(N_B + N_{S_{\text{non-pulsed}}})_{\text{duty cycle}} + N_{S_{\text{pulsed}}}}} \\
 &= \frac{F_x A p_f \Delta t_{\text{obs}}}{\sqrt{[B_X + F_X(1 - p_f)](A \Delta t_{\text{obs}} d) + F_x A p_f \Delta t_{\text{obs}}}} \quad (2.3)
 \end{aligned}$$

- The effective area of the detector was calculated using:

$$A_{\text{eff}}(E) = \epsilon(E) \cdot A_{\text{geom}} \quad (2.4)$$

2.3 Instrument DesignInstrument Design

- Amptek FAST SDD and Preamplifier
- Analog Processing Unit (APU)
- Digital Processing Unit (DPU)
- Power Conditioning Unit (PCU)

2.4 Satellite True Anomaly Estimation:

- Equations to determine the time delay and the true anomaly:

$$\vec{r}_{\text{sat,icrf}} \cdot \vec{n}_{\text{pulsar,icrf}} = c \cdot t_d \quad (2.5)$$

- Step-1: Generating the time delay (td) using the X-ray photon count measurements: The instrument generates time-stamped X-ray photon count values.
- Step-2: Estimate time delay using the obtained t_d :

$$\vec{r}_{\text{sat,icrf}} = \vec{r}_{\text{sat,eci}} - \vec{r}_{\text{SSB, eci}}$$

$$(\vec{r}_{\text{sat,eci}} - \vec{r}_{\text{ssb,eci}}) \cdot \vec{n}_{\text{pulsar,icrf}} = c \cdot t_d$$

$$\vec{r}_{\text{sat,orbit}} = \frac{a \cdot (1 - e^2)}{1 + e \cdot \cos \theta} \cdot \begin{pmatrix} \cos \theta \\ \sin \theta \\ 0 \end{pmatrix} \quad (2.6)$$

Final expression:

$$(Q \cdot \frac{a \cdot (1 - e^2)}{1 + e \cdot \cos \theta} \cdot \begin{pmatrix} \cos \theta \\ \sin \theta \\ 0 \end{pmatrix} - \vec{r}_{\text{SSB,eci}}) \cdot \vec{n} = c \cdot t_d$$

Solving these we'll obtain:

$$\theta = n\pi + (-1)^n \sin^{-1} \left(\frac{f(t)}{\sqrt{k_1^2 + k_2^2}} \right) - K_2$$

where,

$$f(t) = \frac{ct_d - \vec{n} \cdot \vec{r}_{\text{SSB,eci}}}{a \cdot (1 - e^2)}$$

$$K_2 = \sin^{-1} \left(\frac{k_1}{\sqrt{k_1^2 + k_2^2}} \right)$$

Simplified expression:

$$\theta = \begin{cases} \sin^{-1} \frac{f(t)}{\sqrt{k_1^2 + k_2^2}} - K_2 & \text{if } \vec{v}_e \cdot \vec{n} > 0 \\ \pi - \sin^{-1} \frac{f(t)}{\sqrt{k_1^2 + k_2^2}} - K_2 & \text{if } \vec{v}_e \cdot \vec{n} < 0 \end{cases} \quad (2.7)$$

2.5 Simulation and Navigation:

- The simulation uses an orbit propagator from the open-

source Python library Poliastro, which propagates the satellite's orbit in the LEO.

- At different locations in the orbit, the value of t_d was estimated using a reverse-engineered formula for t_d as shown in the equation:

$$t_d = \frac{\vec{r}_{\text{icrf}} \cdot \vec{n}_{\text{icrf}}}{c} \quad (2.8)$$

(In actual flight, this value of t_d would be obtained from the instrument's X-ray photon count measurements)

- The t_d values so obtained were used to calculate the true anomaly using the formula derived in the previous section.
- The results validate the derivation of the formula, as simulated true anomaly and calculated true anomaly match closely, with an average error of less than 1 degree. It is observed that the true anomaly error spikes to around 4 degrees at the locations where the satellite crosses the pulsar vector in the ECI frame.

Chapter 3

Pulse profile generation

Generating pulsar pulse profiles involves processing data from X-ray pulsars to extract their periodic emission patterns. HEASoft, a software package used extensively in high-energy astrophysics, offers tools like XRONOS and PINT that aid in this process. XRONOS facilitates data analysis through pulse folding and phase-resolved spectroscopy, essential for characterizing pulsar emissions. PINT aids in precise timing analysis, aligning observed photon arrival times with a pulsar's rotational phase.[6] Together, these tools enable researchers to construct accurate pulse profiles, crucial for studying pulsar emission mechanisms and navigation applications like X-ray Pulsar-Based Navigation (XNAV) in autonomous spacecraft. [7]

3.1 Downloading Raw NICER Data

- Downloaded HEASoft from HEASARC-HEASoft
- Downloaded data from HEASARC-Data

The primary source of data was the High Energy Astrophysics Science Archive Research Centre's (HEASARC) online Master Catalog, specifically related to the NICER mission. In this catalogue, observations taken on a specific day are grouped and

stored under a unique identifier known as the observation ID. This ID is used to reference the data for any given date.

For pulsar B1937+21, data was filtered to include only observations from NICER between July 1, 2017, and October 31, 2017, when the pulsar was visible and unobstructed. Additionally, only observation IDs with exposure times between 100 and 2000 seconds were included to avoid excessively long observation times and to effectively manage file sizes.

From the resulting table in each case, all observation IDs matching the query constraints were selected, compressed together and downloaded. Once the raw data was downloaded, it was uncompressed into separate directories named according to the pulsar the data corresponds to.

3.2 Calibration of Raw NICER data

3.2.1 Directory Structure

The data set corresponding to each pulsar was arranged according to the NICER/SEXTANT directory structure. This enables HEASoft tools (and also custom-designed, non-HEASoft code) to access the relevant files. Within the directory, files all have prefixes according to the particular observation ID. For example, if the ObsID is 0123456789, the files begin with the prefix ni0123456789 and are suffixed according to which type of file it is. The directory layout is as follows :

```

1200040101
├── auxil
│   ├── nil200040101.att.gz
│   ├── nil200040101.cat
│   ├── nil200040101.mkf.gz
│   └── nil200040101.orb.gz
├── log
│   ├── nil200040101_errlog.html
│   └── nil200040101_joblog.html
└── xti
    ├── event_cl
    │   ├── nil200040101_0mpu7_cl.evt.gz
    │   └── nil200040101_0mpu7_uf.evt.gz
    ├── event_uf
    │   ├── nil200040101_0mpu0_uf.evt.gz
    │   ├── nil200040101_0mpu1_uf.evt.gz
    │   ├── nil200040101_0mpu2_uf.evt.gz
    │   ├── nil200040101_0mpu3_uf.evt.gz
    │   ├── nil200040101_0mpu4_uf.evt.gz
    │   ├── nil200040101_0mpu5_uf.evt.gz
    │   └── nil200040101_0mpu6_uf.evt.gz
    └── hk
        ├── nil200040101_0mpu0.hk.gz
        ├── nil200040101_0mpu1.hk.gz
        ├── nil200040101_0mpu2.hk.gz
        ├── nil200040101_0mpu3.hk.gz
        ├── nil200040101_0mpu4.hk.gz
        ├── nil200040101_0mpu5.hk.gz
        └── nil200040101_0mpu6.hk.gz

```

Figure 3.1: An example of the file arrangement according to the NICER/-SEXTANT directory structure. Note that this is the complete set of data files made available for any ObsID, and for this project’s purposes, the xti/ and auxil/ folders are only required from the NICER Mission Guide.

3.2.2 Calibration done using NICER’s caldb

Calibration data was required to produce scientific results using the NICER analysis software. Like many other missions, NICER utilizes the HEASARC’s Calibration Database (CALDB) system to catalogue and distribute calibration data.

The CALDB system not only archives and distributes calibration data but also tags each data set with a unique release identifier. This tagging capability allows analysts to describe the source of their calibration data in their papers.

To install a local copy of CALDB, a ”framework” was first downloaded and installed, followed by the NICER-specific files.

3.2.3 Retrieving and Using Geomagnetic Quantities

NICER is a non-imaging instrument that collects X-rays through concentrator optics into a silicon drift diode detector. This detector functions as a single-pixel device or "light bucket," capturing every X-ray event concentrated by the optics.

In addition to X-rays, the detectors also collect background events, many of which are due to NICER's space environment, which is the same as that of the International Space Station (the host of NICER). NICER's orbit takes it through radiation belts and the South Atlantic Anomaly. The detectors may also capture solar or extra-solar cosmic rays. These background events are typically considered nuisance events that need to be modeled.

One current background model utilizes a "Space Weather" method to estimate the background. Detailed information about the Space Weather model can be found on the NICER Background Estimators page. A key input to this model is the state of the space weather environment, characterized by the Kp-index.

Another model is the SCORPEON background model, which uses geomagnetic data quantities. Specifically, it utilizes the solar modulation potential (SOLAR PHI) to estimate the modulation of cosmic-ray-related background, which may vary slowly over time.[8]

3.3 Data Processing Pipeline

To help automate this process, a wrapper script (`amoy-spectral-analysis.sh`) was written to carry out the processes and is provided in the appendix. The tasks involved are "nicerl2" and "nicerl3-spect" and the plotting commands for XSPEC.

3.3.1 Basic calibration

The task `nicerl2` performs the NICER team-recommended calibration and screening steps for NICER data.

The following tasks can also be run instead:

- `nicercal` (Apply standard calibration to NICER observation)
- `niprefilterl2` (Create an augmented NICER-specific filter file; this additional information is more useful for filtering and generating background estimates)
- `nimaketime` (Used for creating a good time interval file [These files are used to screen for photon events that have the following attributes: The ISS is not within the NICER-specific definition of the South Atlantic Anomaly (SAA) region. NICER is tracking the source. NICER is pointing within 0.015° of the source. The source is at least 30° of the Earth's limb. Only photon arrivals lying within the GTIs are carried forth.])

3.3.2 Spectral analysis

Spectroscopy is essential for most X-ray data analysis projects, as it provides unique information about X-ray targets that cannot be obtained by other means. Therefore, extracting a spectrum is a crucial task for nearly all NICER users.

This process involves using a single pipeline task, `nicerl3-spect`, which extracts all NICER spectral products. The task generates all necessary products for spectral analysis, including the spectrum, background estimate, response files (ARF and RMF), and additional script files to facilitate analysis.

The `nicerl3-spect` task was introduced with the release of HEASoft 6.31 in November 2022. It offers a straightforward

and efficient method to extract spectral products using NICER-recommended techniques.

The `nicerl3-spect` task will make many automatic decisions for you, such as choice of background model, but the advanced user is free to make alternate choices.

As output, `nicerl3-spect` makes the following products:

- extracted spectrum from cleaned event file (`phafile`)
- Background spectrum file (`bkgfile`) or script (`bkgscript`) depending on the background model setting
- ARF response file (`arffile`) - effective area for target
- Sky ARF response file (`skyarffile`) - effective area for diffuse sky
- RMF response file (`rmffile`) - redistribution matrix for target
- Background response file (`bkgmffile`) - redistribution matrix for particle background
- XSPEC "load" file (`loadfile`) - example script to load files into XSPEC

3.4 Generating light curves

To help automate this process, a wrapper script (`amoy-light-curves.sh`) was written to carry out the processes and is provided in the appendix. The task involved is "nicerl3-lc".

Light curve analysis plays a crucial role in X-ray data analysis, providing essential context in many scenarios. For instance, it helps detect bursts or eclipses, allowing scientists to pinpoint significant events. Moreover, light curve analysis serves as the foundation for timing analysis, including power spectroscopy.

As a result, extracting a light curve is a fundamental task that virtually all NICER users will find necessary.

This guide details the extraction of all NICER light curve products using a single pipeline task known as `nicerl3-lc`. This task efficiently generates all required products for light curve analysis, including the light curve itself and an optional background estimate.

The `nicerl3-lc` task was introduced with the release of HEASoft 6.31 in November 2022, offering a straightforward and efficient method to extract these products using NICER recommended methodologies.

3.5 Generating Power Spectrum

To help automate this process, a wrapper script (`amoy-powerspectrum.sh`) was written to carry out the processes and is provided in the appendix. The tasks involved are "nicerl3-lc" and the commands required for generating the power spectrum in "powspec".

Power spectrum analysis is a fundamental technique in X-ray astronomy and signal processing. It involves decomposing a time series into its frequency components to analyze periodicities or variability in data. In the context of NICER data analysis, power spectrum analysis is crucial for studying pulsar signals, investigating variability in X-ray sources, and understanding the underlying physical processes driving these variations. By extracting the power spectrum, scientists can identify periodic signals, measure their strength at different frequencies, and infer properties such as rotation periods of pulsars or variability timescales of astrophysical phenomena. The process typically involves Fourier transforming the time-domain data into the frequency domain, where the resulting power spectrum reveals the distribution of signal power across different frequencies, pro-

viding insights into the dynamics and characteristics of celestial objects observed by NICER.

3.6 Pulse Profile Generation

To help automate this process, a wrapper script (`amoy-efold.sh`) was written to carry out the processes and is provided in the appendix. The task involved is "efold" and the plotting commands required for generating the epoch-folded pulse profile.

Epoch folding is a technique used in X-ray astronomy to analyze periodic signals from celestial sources, particularly pulsars. It involves aligning and folding multiple observations of an astronomical object based on its known or predicted period. By phase-aligning these observations, epoch folding enhances the signal-to-noise ratio in the resulting folded light curve, thereby revealing the underlying periodicity more clearly. This method is essential for studying the rotational periods of pulsars and other periodic phenomena in X-ray sources observed by instruments like NICER. Epoch folding allows astronomers to extract precise timing information, measure spin periods, detect changes in pulsar rotational behavior over time, and study variations in X-ray emission patterns. It is a foundational technique in pulsar astronomy, enabling detailed investigations into the physics of neutron stars and other astrophysical objects emitting periodic X-ray signals.

3.7 Results

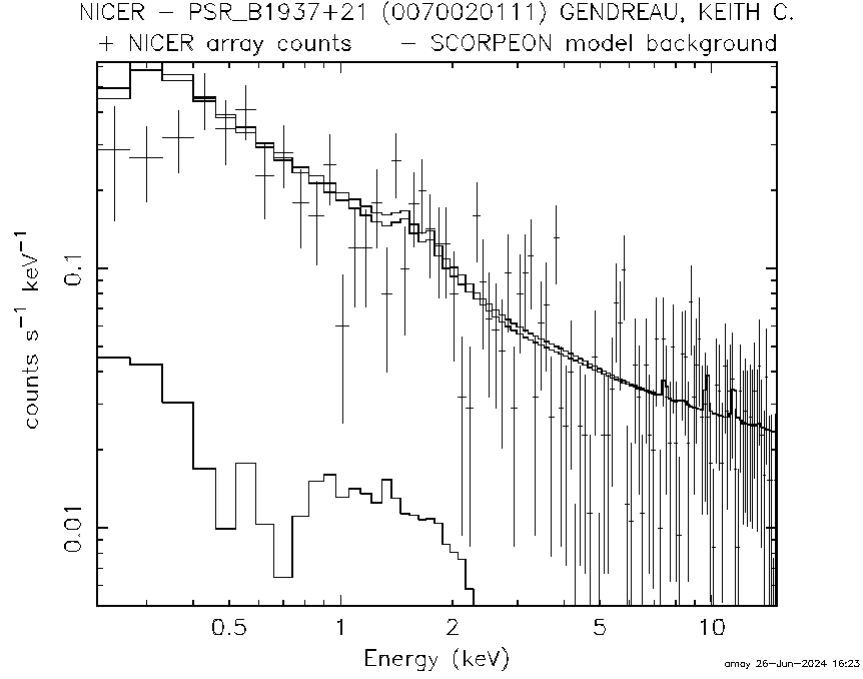


Figure 3.2: Example XSPEC output. In this example, we have used the default SCORPEON background model which is an XSPEC model. This model is shown with the data (i.e. no background subtraction)

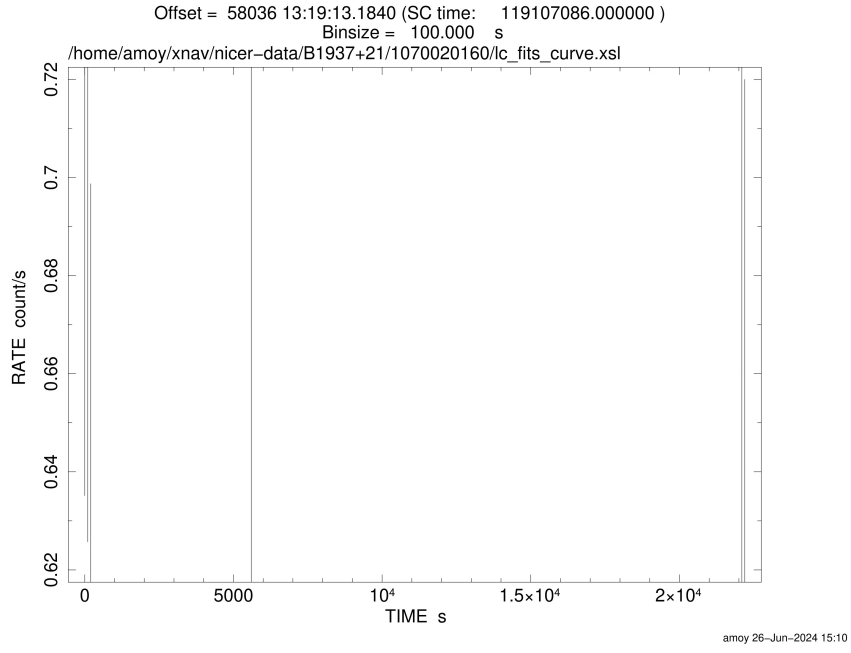


Figure 3.3: Example nicerl3-lc light curve plot.

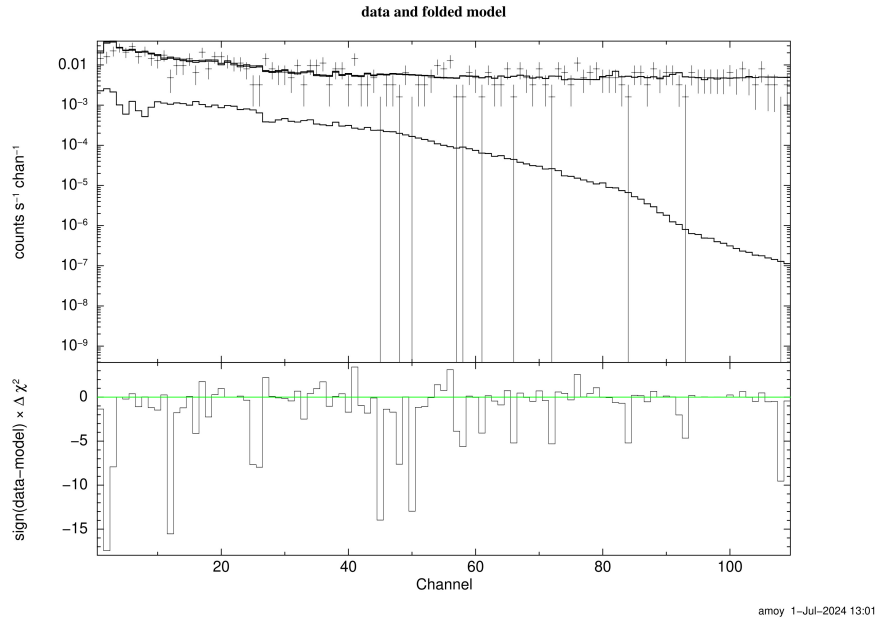


Figure 3.4: Example powerlaw output.

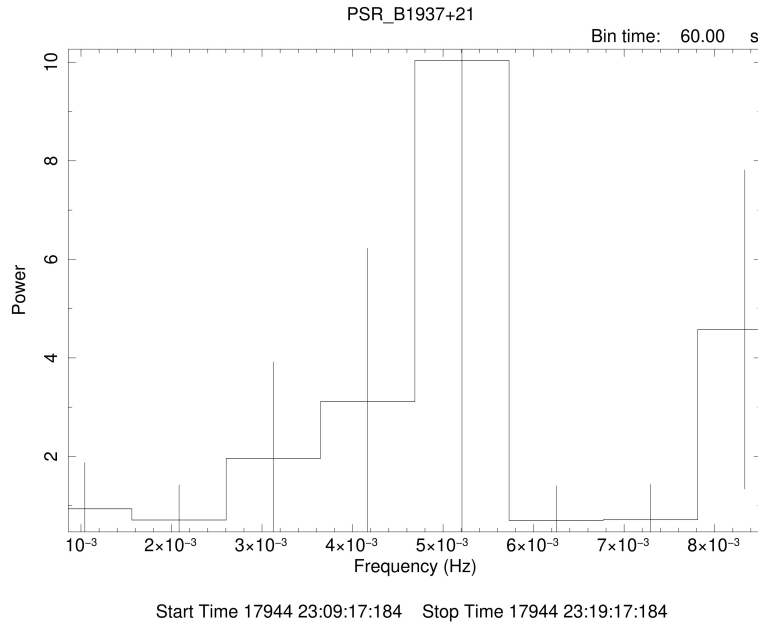


Figure 3.5: Example power spectrum.

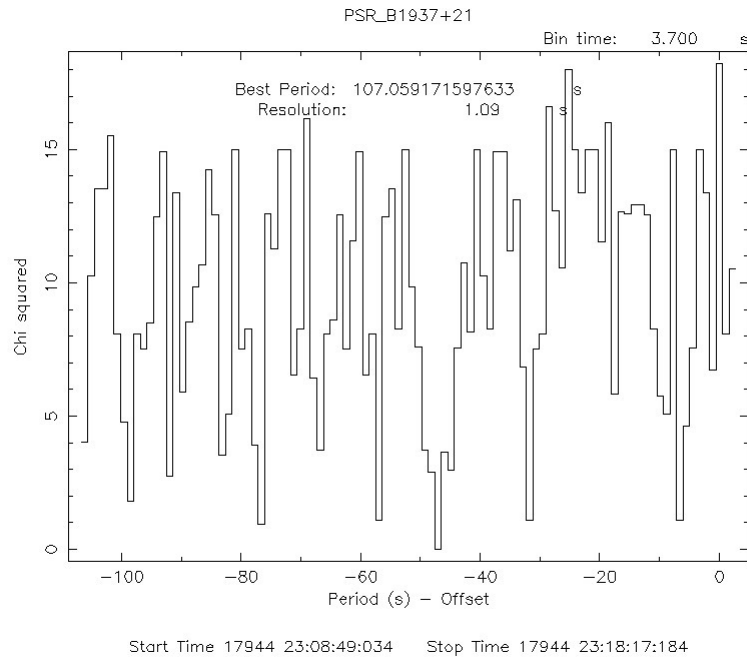


Figure 3.6: Example powspec Output

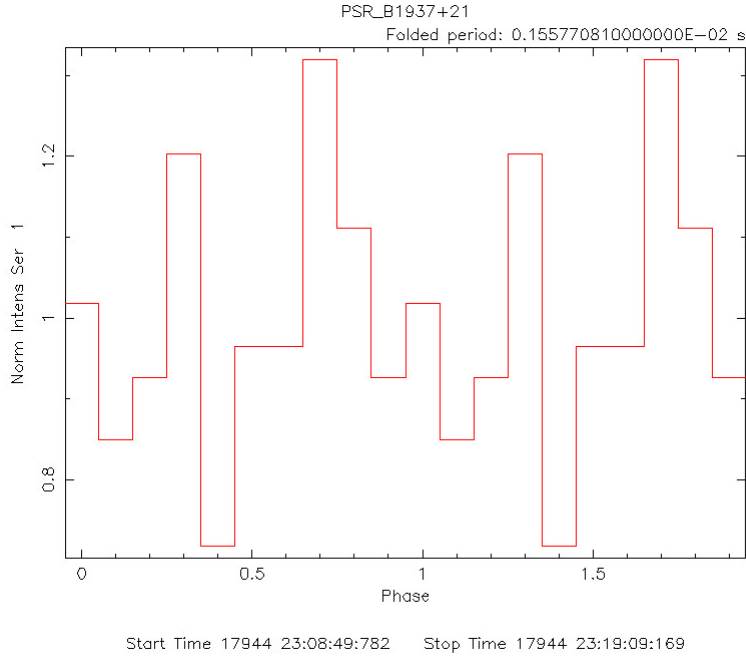


Figure 3.7: Example pulsar pulse profile

3.8 Conclusions

In this report, we have explored the capabilities of HEAsoft in analyzing and plotting various astronomical data products, including power spectra, pulsar pulse profiles, epoch folded profiles, and light curves. HEAsoft provides a versatile suite of tools that are essential for studying X-ray sources observed by instruments like NICER. Below, we summarize the methodologies employed, key findings, and implications of our analyses.

3.8.1 Power Spectrum Analysis

Power spectrum analysis is crucial for understanding the periodic and stochastic variability of X-ray sources. Using tools like `powspec` and `powspec2`, we demonstrated how HEAsoft enables the extraction of power spectra from time-series data. The power spectra reveal characteristic frequencies and variability timescales,

essential for studying pulsar timing, accretion processes, and other astrophysical phenomena. Our analyses underscored the importance of Fourier techniques in uncovering periodic signals and measuring their strength across different frequencies.

3.8.2 Pulsar Pulse Profiles

HEAsoft facilitates the generation and visualization of pulsar pulse profiles using tools like `nicerl3-prof`. These profiles provide insights into the emission patterns and rotational characteristics of pulsars. By folding photon arrival times with the pulsar period, we obtained phase-resolved pulse profiles that highlight emission peaks and variations over the pulsar’s rotational cycle. Such profiles are vital for studying pulsar emission mechanisms, magnetic field configurations, and emission geometry.

3.8.3 Epoch Folded Profiles

Epoch folding techniques in HEAsoft allow for the coherent stacking of multiple observations to enhance the detection of periodic signals. Using tools such as `efold` and `efsearch`, we demonstrated how epoch folding improves the signal-to-noise ratio in folded light curves. This method is instrumental in detecting subtle changes in periodic behavior over time, providing deeper insights into pulsar spin evolution, binary orbital dynamics, and other time-variable phenomena observed by X-ray telescopes.

3.8.4 Light Curves

Light curve analysis using HEAsoft, facilitated by tools like `nicerl3-lc`, offers a comprehensive view of X-ray intensity variations over time. By extracting and plotting light curves, we identified flux variations, bursts, eclipses, and other transient events in X-ray sources. These analyses are essential for studying

the temporal behaviour of astrophysical objects, characterizing variability timescales, and investigating the underlying physical processes driving X-ray emission.

Chapter 4

Time Synchronization using Pulsars

4.1 BASIC OUTLINE

4.1.1 Processing the Pulsar Signal:

To extract a reliable timing reference, scientists employ advanced signal processing techniques:[9]

1. De-dispersion: Special algorithms mathematically correct for the interstellar dispersion, recovering the original pulse shape.
2. Folding: The received signal is segmented and "folded" based on the pulsar's known rotation period. This process aligns the individual pulses, creating a composite profile with a clear peak representing the precise arrival time.

4.1.2 Timekeeping with Pulsars:

- By meticulously measuring the arrival times of pulsar signals, astronomers can accurately determine the pulsar's rotational period.

- Any deviations in the arrival times can be attributed to variations in the Earth’s rotation or the observer’s position.
- This information is used to refine terrestrial time standards like International Atomic Time (TAI), ensuring the utmost precision.

4.1.3 Challenges and Solutions

- **Pulse Jitter:** Pulse arrival times can vary slightly due to interstellar medium effects or intrinsic pulsar properties. This jitter can be mitigated by averaging multiple pulses.
- **Observational Cadence:** Frequent and regular observations are necessary to maintain accurate timing models. Networks of telescopes worldwide help achieve continuous monitoring.
- **Interstellar Medium (ISM):** The ISM can affect pulse arrival times through dispersion and scattering. Correcting for these effects involves modeling the ISM and using multi-frequency observations.

4.1.4 Advanced Techniques

- **Frequency Drift Analysis:** The slight changes in pulsar frequency over time (due to rotational slowdown or glitches) are monitored and modeled.
- **Relativistic Corrections:** Precise models incorporate corrections for relativistic effects, such as the Shapiro delay (caused by gravitational fields bending the pulse path).
- **Global Timing Efforts:** International collaborations, like the International Pulsar Timing Array (IPTA), pool data from different telescopes to improve timing accuracy and create a comprehensive pulsar-based timekeeping system.

4.2 NAVIGATION OVERVIEW

4.2.1 X-ray Navigation System (XNAV):

XNAV is a spacecraft navigation system concept that utilizes the precise timing of MSP signals. Here's the breakdown: Onboard Detectors: The spacecraft carries specialized X-ray detectors capable of accurately registering the arrival times of MSP pulses.

- **Pulsar Catalogues:** Detailed catalogues of MSPs are essential. These catalogues include their positions, rotation periods, and other relevant information.
- **Timing and Calculations:** The spacecraft compares the arrival times of known MSP pulses with the predicted arrival times based on the catalogue data. Deviations in arrival times reveal information about the spacecraft's position and velocity.

4.2.2 The Navigation Process:

The navigation process involves a series of steps:

- **Target Selection:** The spacecraft identifies suitable MSPs for navigation based on factors like signal strength and geometric positioning.
- **Pulse Detection and Timing:** The X-ray detectors capture the MSP pulses, and their arrival times are meticulously recorded with onboard clocks.
- **Data Analysis:** Sophisticated algorithms analyze the recorded arrival times and compare them with the predicted arrival times based on the catalog data and the spacecraft's initial estimates of its position and velocity.

- Position and Velocity Determination: Discrepancies between the observed and predicted arrival times allow the system to accurately calculate the spacecraft's position and velocity.
- Iterative Refinement: As the spacecraft navigates, it continuously collects and analyzes pulsar data, refining its positional and velocity estimates over time.

Chapter 5

Collimator design calculations

5.1 Conical design

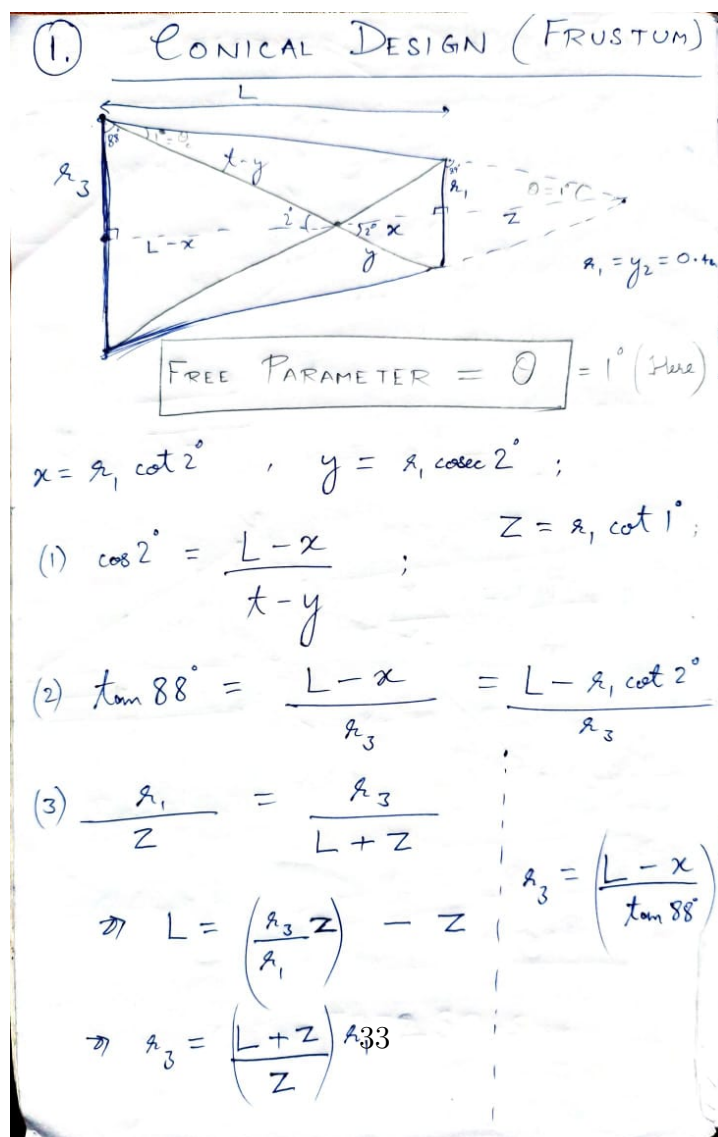


Figure 5.1: Part 1

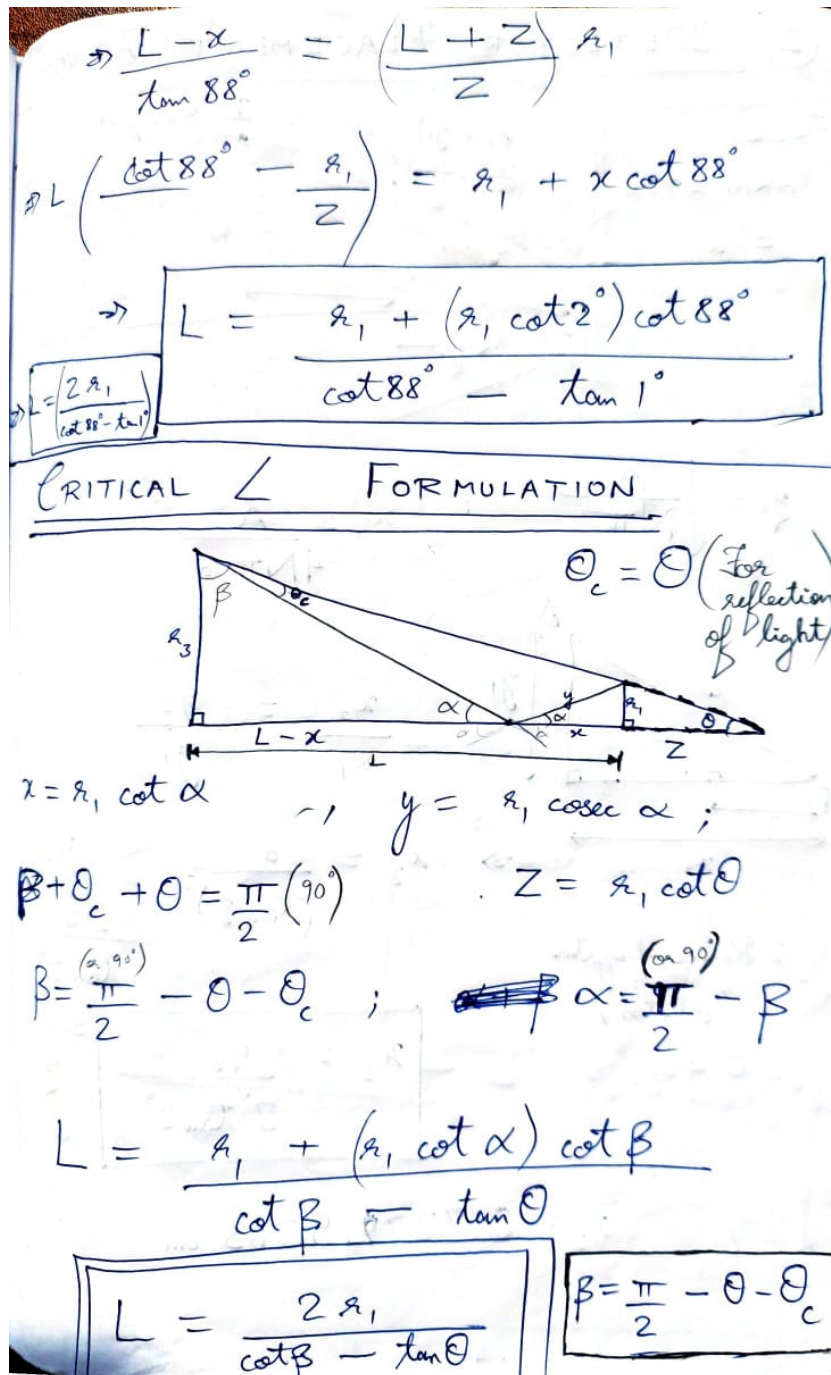


Figure 5.2: Part 2

5.2 Paraboloid design

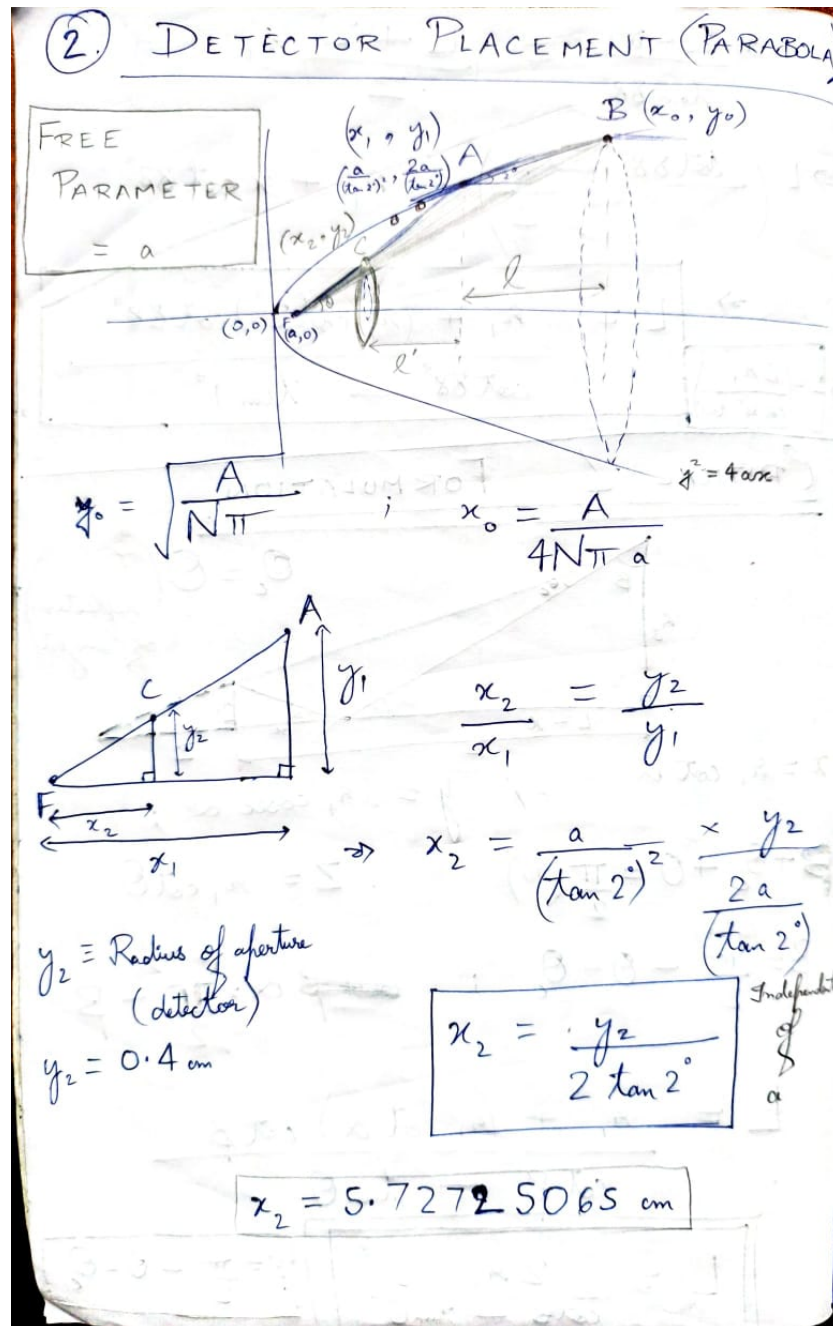


Figure 5.3: Part 1

$$\tan \theta = \frac{y_1}{x_1 - a} = \frac{\left(\frac{2a}{\tan 2^\circ}\right)}{\left(\frac{a}{(\tan 2^\circ)^2}\right) - a} = \frac{2(\tan 2^\circ)}{1 - (\tan 2^\circ)^2}$$

$$\rightarrow \tan \theta = 0.069841538983495$$

$$\rightarrow \theta = 3.975138^\circ \quad (\text{Independent of } a)$$

$$\boxed{\theta = 4^\circ} \quad \theta = \tan^{-1}[\tan(2\theta_c)] = 2\theta_c$$

$2^\circ \rightarrow$ Call it a "critical" $\angle$ for X-rays $\equiv \theta_c$

CRITICAL $\angle$ formulation

$$x_0 = \frac{A}{4\pi Na}, \quad y_0 = \sqrt{\frac{A}{N\pi}}$$

$$x_1 = \frac{a}{(\tan \theta_c)^2}, \quad y_1 = \frac{2a}{(\tan \theta_c)}$$

$$\boxed{x_2 = \frac{y_2}{2 \tan \theta_c}}$$

$$\boxed{\theta = \tan^{-1} \left[\frac{2(\tan \theta_c)}{1 - (\tan \theta_c)^2} \right]}$$

$$\boxed{\theta = 2\theta_c}$$

Effective length of paraboloid $\equiv l = \frac{A}{4\pi Na} - \frac{a}{(\tan \theta_c)^2}$

$$\boxed{l' = \left(\frac{a}{\tan \theta_c} - \frac{y_2}{2 \tan \theta} \right)}$$

Figure 5.4: Part 2

Chapter 6

Summary & Future Work

6.1 Summary

This report lays the groundwork for a complete X-ray pulsar navigation (XNAV) system. It explores the theory behind XNAV, signal processing techniques, and even a basic pulse processing circuit design. While the report focuses on the core concepts and doesn't explore complex technical details, it provides a valuable foundation for building an XNAV system for an Earth-orbiting satellite.

A key part of the report is the development of an automated system to generate and analyze the special X-ray signals emitted by pulsars, called pulse profiles. This system, written mostly in shell scripts, uses existing tools like HEASoft to create detailed and accurate pulse profiles. Scripts written in Bash handle the automation and data organization, ensuring a smooth process.

Pulse profiles are essential for XNAV because they allow scientists to translate the received X-ray signals from pulsars into usable information for navigation. By considering the known rotation period of a pulsar, the arrival times of X-ray photons are folded into a profile, revealing the highs and lows of the pulsar's emission. These profiles are vital for identifying and understanding the pulsar's signal, which ultimately helps refine

navigation models and improve accuracy.

The automation developed in this project ensures consistent and reliable pulse profiles, minimizing errors and streamlining data processing. This is crucial for real-time navigation, where speed and precision are paramount. Additionally, the Bash script framework seamlessly integrates HEASoft, creating a robust and adaptable platform for ongoing XNAV development.

The successful testing of this pulse processing code demonstrates its ability to generate high-quality data, paving the way for further XNAV system development. The automation and software integration marks a significant step forward for X-ray navigation.

In essence, this report represents a major leap in XNAV technology by automating pulse profile generation for millisecond pulsars. The established techniques and tools provide a strong base for future research, ultimately contributing to reliable and accurate X-ray navigation for spacecraft. The combination of Python with HEASoft and PINT opens doors for even more advanced and effective XNAV systems in the future.

6.2 Future work

This report has laid the groundwork for developing an X-ray Pulsar Navigation (XNAV) system, but several key areas require further research and exploration to fully realize a robust and effective navigation solution. Future work should address the following questions and challenges:

1. **Occultations from Other Sources Inside the Solar System:** Investigate the impact of occultations caused by objects within the solar system, such as planets, moons, and asteroids. Develop strategies to predict and mitigate these events to ensure continuous and accurate navigation.

2. **Occultations from Unknown Deep Space Sources:** Explore how deep space missions can account for occultations from unknown sources, including black holes, gas clouds, and other interstellar objects. This includes developing detection and avoidance strategies to maintain reliable navigation.
3. **Satellite Navigation During Occultations:** Determine methods for the satellite to navigate effectively during occultations. This might involve using alternative navigation aids or predictive modeling to maintain course and timing accuracy when pulsar signals are temporarily obscured.
4. **Locking Onto the Pulsar Under Consideration:** This report has addressed how the satellite can lock onto the pulsar under consideration. Observations are conducted for specific periods, and onboard instruments and algorithms are capable of precisely pointing the spacecraft in the desired direction. This process ensures accurate and consistent data collection from targeted pulsars.
5. **Eliminating Other Periodic X-ray Sources:** Develop techniques to differentiate between the targeted pulsar signals and other periodic X-ray sources, such as Cepheid variables or nearby pulsars. This will involve refining signal processing algorithms to filter out unwanted noise and interference, ensuring that only the desired pulsar signals are used for navigation.
6. **Multiple Pulsars in the Observational Area:** Address the challenge of multiple pulsars present within the same observational area, especially when they are radially separated. This involves creating strategies to identify and isolate individual pulsar signals, allowing for accurate navigation using

the correct pulsar sources without confusion from adjacent signals.

These future work directions outline the next steps needed to enhance the capabilities and reliability of XNAV systems. By addressing these challenges, we can move closer to achieving a fully operational and dependable navigation system based on X-ray pulsar signals.

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Appendix A

Pulsar Processing Codes

A.1 Main Bash Scripts

```
1  #!/bin/bash
2  #-----
3  #-----
4  ### Define user-paths for convenience
5
6  export DATAFOLDER=/home/amoy/xnav/nicer-data/B1937+21
7  export CODESFOLDER=/home/amoy/xnav/codes
8
9  #-----
10 #-----
11 ### Loop over the ObsIds in the data set,
12 ### and merge the various files required for
13 ### making the pulse profile
14
15 ls $DATAFOLDER > $CODESFOLDER/obsids.txt
16 mapfile -t < $CODESFOLDER/obsids.txt # get list of
    ObsIDs as array
17
18
19 for i in "${MAPFILE[@]}"
20 do
21     cd "$DATAFOLDER" || exit
22     export CWD=$DATAFOLDER/$i # Current Working
        Directory
23     export CURAUX=$CWD/auxil # Current auxil Directory
24     export CURXTI=$CWD/xti # Current xti Directory
25
```

```

26  echo
27      echo "-----"
28      echo "*****Looping ... Currently processing ObsID
        $i*****"
29      echo "-----"
30
31  nicerl2 indir=$i clobber=YES
32  nicerl3-spect indir=$i clobber=YES
33  echo
34
35  # Find the _load.xcm file
36  LOAD_XCM=$(find "$CURXTI/event_cl" -type f -name
        "*_load.xcm")
37
38      if [[ -f "$LOAD_XCM" ]]; then
39          xspec <<EOF
40  @$LOAD_XCM
41  cd $i
42  cpd spect-$i/ps
43  setplot energy
44  setplot rebin 1 2
45  plot ldata
46  cpd none
47  exit
48  EOF
49      else
50          echo "File *_load.xcm not found for ObsID $i"
51      fi
52
53  done

```

Listing A.1: amoy-spectral-analysis.sh

```

1  #!/bin/bash
2  #-----
3  #-----
4  ### Define user-paths for convenience
5
6  export DATAFOLDER=/home/amoy/xnav/nicer-data/B1937+21
7  export CODESFOLDER=/home/amoy/xnav/codes
8
9  #-----
10 #-----
11 ### Loop over the ObsIds in the data set,

```

```

12 ### and merge the various files required for
13 ### making the pulse profile
14
15 ls $DATAFOLDER > $CODESFOLDER/obsids.txt
16 mapfile -t < $CODESFOLDER/obsids.txt # get list of
   ObsIDs as array
17
18
19 for i in "${MAPFILE[@]}"
20 do
21   cd "$DATAFOLDER" || exit
22   export CWD=$DATAFOLDER/$i # Current Working
   Directory
23   export CURAUX=$CWD/auxil # Current auxil Directory
24   export CURXTI=$CWD/xti # Current xti Directory
25
26   echo
27   echo "-----"
28   echo "*****Looping ... Currently processing ObsID
   $i*****"
29   echo "-----"
30
31   echo
32   nicerl3-lc indir=$i pirange=300-1500 timebin=1E-4
   clobber=YES
33
34 done

```

Listing A.2: amoy-light-curves.sh

```

1 #!/bin/bash
2 #-----
3 #-----
4 ### Define user-paths for convenience
5
6 export DATAFOLDER=/home/amoy/xnav/nicer-data/B1937+21
7 export CODESFOLDER=/home/amoy/xnav/codes
8
9 #-----
10 #-----
11 ### Loop over the ObsIds in the data set,
12 ### and merge the various files required for
13 ### making the pulse profile
14

```

```

15 mapfile -t < $CODESFOLDER/obsids.txt # get list of
    ObsIDs as array
16
17
18 for i in "${MAPFILE[@]}"
19 do
20     cd $DATAFOLDER
21     export CWD=$DATAFOLDER/$i # Current Working
        Directory
22     export CURAUX=$CWD/auxil # Current auxil Directory
23     export CURXTI=$CWD/xti # Current xti Directory
24
25     echo
26     echo "-----"
27     echo "*****Looping ... Currently processing ObsID
        $i*****"
28     echo "-----"
29     echo
30
31     nicerl3-lc $i pirange=300-1500 timebin=60.0
        bkgmodeltype=scorpeon clobber=YES
32
33     LOAD_LC=$(find "$CURXTI/event_cl" -type f -name
        "*_sr.lc")
34     LC_FILE=$(find "$CURXTI/event_cl" -type f -name
        "*_sr.lc" -exec basename {} \; | tr '\n' ' ')
35     if [[ -f "$LOAD_LC" ]]; then
36         cd $CURXTI/event_cl
37         powspec <<EOF
38 $LC_FILE
39 -
40 INDEF
41 INDEF
42 INDEF
43 0
44 default
45 yes
46 /XW
47 color 2 on 2
48 cpd $i.gif/gif
49 p
50 quit
51 EOF

```



```

52     else
53         echo "File *_sr.lc not found for ObsID $i"
54     fi
55 done

```

Listing A.3: amoy-powerspectrum.sh

```

1  #!/bin/bash
2  #-----
3  #-----
4  ### Define user-paths for convenience
5
6  export DATAFOLDER=/home/amoy/xnav/nicer-data/B1937+21
7  export CODESFOLDER=/home/amoy/xnav/codes
8
9  #-----
10 #-----
11 ### Loop over the ObsIds in the data set,
12 ### and merge the various files required for
13 ### making the pulse profile
14
15 mapfile -t < $CODESFOLDER/obsids.txt # get list of
    ObsIDs as array
16
17
18 for i in "${MAPFILE[@]}"
19 do
20     cd $DATAFOLDER
21     export CWD=$DATAFOLDER/$i # Current Working
        Directory
22     export CURAUX=$CWD/auxil # Current auxil Directory
23     export CURXTI=$CWD/xti # Current xti Directory
24
25     echo
26     echo "-----"
27     echo "*****Looping ... Currently processing ObsID
        $i*****"
28     echo "-----"
29     echo
30
31     LOAD_LC=$(find "$CURXTI/event_cl" -type f -name
        "*_sr.lc")
32     LC_FILE=$(find "$CURXTI/event_cl" -type f -name
        "*_sr.lc" -exec basename {} \; | tr '\n' ' ')

```

```

33  if [[ -f "$LOAD_LC" ]]; then
34      cd $CURXTI/event_cl
35      efold nser=1 cfile1="$LC_FILE" window="-"
          seepoch=INDEF dper=1.5577081E-3 nphase=10
          nbin=3194 nintfm=INDEF plot=yes plotdev="/xw"
          plotdnum=1 outfile="-" <<EOF
36 line step
37 color 5 on 2
38 error off
39 cpd folded_${i}.gif/gif
40 p
41 quit
42 EOF
43 else
44     echo "File *_sr.lc not found for ObsID $i"
45     fi
46 done

```

Listing A.4: amoy-efold.sh

A.2 Auxil Bash scripts

```

1  #!/bin/bash
2  #-----
3  #-----
4  ### Define user-paths for convenience
5
6  export DATAFOLDER=/home/amoy/xnav/nicer-data/B1937+21
7  export CODESFOLDER=/home/amoy/xnav/codes
8
9  #-----
10 #-----
11 ### Loop over the ObsIds in the data set,
12 ### and merge the various files required for
13 ### making the pulse profile
14
15 mapfile -t < $CODESFOLDER/obsids.txt # get list of
    ObsIDs as array
16
17
18 for i in "${MAPFILE[@]}"
19 do

```

```

20 cd $DATAFOLDER
21     export CWD=$DATAFOLDER/$i # Current Working
        Directory
22     export CURAUX=$CWD/auxil # Current auxil Directory
23     export CURXTI=$CWD/xti # Current xti Directory
24
25 echo
26     echo "-----"
27     echo "*****Looping ... Currently processing ObsID
        $i*****"
28     echo "-----"
29 echo
30
31 LOAD_CL=$(find "$CURXTI/event_cl" -type f -name
        "*_cl.evt")
32 CL_FILE=$(find "$CURXTI/event_cl" -type f -name
        "*_cl.evt" -exec basename {} \; | tr '\n' ' ')
33 ORB_FILE=$(find "$CURAUX" -type f -name "*.orb" )
34
35 if [[ -f "$LOAD_CL" ]]; then
36     cd $CURXTI/event_cl
37     chmod +rwx $CL_FILE
38     OUTFILE="$${i}_bary.evt"
39     barycorr infile=$CL_FILE outfile=$OUTFILE
        orbitfiles=$ORB_FILE ra=38.560210
        dec=59.14166 barytime=yes
40 else
41     echo "File *_cl.evt not found for ObsID $i"
42 fi
43 done

```

Listing A.5: amoy-barytime.sh

```

1 #!/bin/bash
2 #-----
3 #-----
4 ### Define user-paths for convenience
5
6 export DATAFOLDER=/home/amoy/xnav/nicer-data/B1937+21
7 export CODESFOLDER=/home/amoy/xnav/codes
8
9 #-----
10 #-----
11 ### Loop over the ObsIds in the data set,

```

```

12 ### and merge the various files required for
13 ### making the pulse profile
14
15 mapfile -t < $CODESFOLDER/obsids.txt # get list of
   ObsIDs as array
16
17
18 for i in "${MAPFILE[@]}"
19 do
20     cd $DATAFOLDER
21     export CWD=$DATAFOLDER/$i # Current Working
   Directory
22     export CURAUX=$CWD/auxil # Current auxil Directory
23     export CURXTI=$CWD/xti # Current xti Directory
24
25     echo
26     echo "-----"
27     echo "*****Looping ... Currently processing ObsID
   $i*****"
28     echo "-----"
29     echo
30
31     LOAD_LC=$(find "$CURXTI/event_cl" -type f -name
   *_sr.lc")
32     LC_FILE=$(find "$CURXTI/event_cl" -type f -name
   *_sr.lc -exec basename {} \; | tr '\n' ' ')
33     if [[ -f "$LOAD_LC" ]]; then
34         cd $CURXTI/event_cl
35         efsearch cfile1="$LC_FILE" window="-" seepoch=INDEF
   dper=37 nphase=10 nbint=INDEF nper=100
   dres=INDEF outfile="-" plot=yes plotdev="/xw"
   <<EOF
36     cpd $i.gif/gif
37     p
38     quit
39     quit
40     EOF
41     else
42         echo "File *_sr.lc not found for ObsID $i"
43     fi
44 done

```

Listing A.6: amoy-efsearch.sh

```

1  #!/bin/bash
2  #-----
3  #-----
4  ### Define user-paths for convenience
5
6  export DATAFOLDER=/home/amoy/xnav/nicer-data/B1937+21
7  export CODESFOLDER=/home/amoy/xnav/codes
8
9  #-----
10 #-----
11 ### Loop over the ObsIds in the data set,
12 ### and merge the various files required for
13 ### making the pulse profile
14
15 mapfile -t < $CODESFOLDER/obsids.txt # get list of
    ObsIDs as array
16
17
18 for i in "${MAPFILE[@]}"
19 do
20     cd $DATAFOLDER
21     export CWD=$DATAFOLDER/$i # Current Working
        Directory
22     export CURAUX=$CWD/auxil # Current auxil Directory
23     export CURXTI=$CWD/xti # Current xti Directory
24
25     echo
26     echo "-----"
27     echo "*****Looping ... Currently processing ObsID
        $i*****"
28     echo "-----"
29     echo
30
31     # Find the _load.xcm file
32     LOAD_CL=$(find "$CURXTI/event_cl" -type f -name
        "*_cl.evt")
33     CL_FILE=$(find "$CURXTI/event_cl" -type f -name
        "*_cl.evt" -exec basename {} \; | tr '\n' ' ')
34
35     if [[ -f "$LOAD_CL" ]]; then
36         cd $CWD
37         xselect <<EOF

```

```

38 lc
39 read events
40 xti/event_cl
41 $CL_FILE
42 yes
43 set binsize 100
44 extract curve
45 plot curve
46 hardcopy /PS
47 exit
48 save curve lc0$i.fits
49 exit
50 no
51 EOF
52     else
53         echo "File *_cl.evt not found for ObsID $i"
54     fi
55
56 done

```

Listing A.7: amoy-fits-maker.sh

```

1  #!/bin/bash
2  #-----
3  #-----
4  ### Define user-paths for convenience
5
6  export DATAFOLDER=/home/amoy/xnav/nicer-data/B1937+21
7  export CODESFOLDER=/home/amoy/xnav/codes
8
9  #-----
10 #-----
11 ### Loop over the ObsIds in the data set,
12 ### and merge the various files required for
13 ### making the pulse profile
14
15 mapfile -t < $CODESFOLDER/obsids.txt # get list of
    ObsIDs as array
16
17
18 for i in "${MAPFILE[@]}"
19 do
20     cd $DATAFOLDER
21     export CWD=$DATAFOLDER/$i # Current Working

```

```

22         Directory
23     export CURAUX=$CWD/auxil # Current auxil Directory
24
25     echo
26     echo "-----"
27     echo "*****Looping ... Currently processing ObsID
28         $i*****"
29     echo "-----"
30
31     # Find the _load.xcm file
32     LOAD_XCM=$(find "$CURXTI/event_cl" -type f -name
33         "*_load.xcm")
34
35     if [[ -f "$LOAD_XCM" ]]; then
36         xspec <<EOF
37     @$LOAD_XCM
38     cd $i
39     model phabs(powerlaw)
40     /*
41     renorm
42     ignore bad
43     cpd powerlaw-$i.ps/cps
44     plot ldata chi
45     fit
46     y
47     y
48     y
49     n
50     plot data resid
51     cpd none
52     exit
53     EOF
54     else
55         echo "File *_load.xcm not found for ObsID $i"
56     fi
57
58 done

```

Listing A.8: amoy-powerlaw.sh